

XTechon – Full- Stack Developer Technical Assignment

Project: Flight Booking System (End-to-End)

This assignment is designed to evaluate your full-stack development skills, code quality, architectural thinking, and ability to implement real-world features. Please ensure your submission reflects professional, production-ready practices.

1. Flight Search Module (Database Required)

You must build a database-driven flight search system. Static JSON, random generation, or external APIs are strictly prohibited.

- Requirements:
- Seed 10–20 flights into your database (MongoDB / MySQL / PostgreSQL).
- Each flight must include: flight_id, airline, departure_city, arrival_city, base_price (₹2000–₹3000).
- Every search must return 10 flights fetched directly from the database.

2. Dynamic Pricing Engine

Implement surge pricing based on booking attempts within defined time windows.

- Rules:
- If a user tries to book the same flight 3 times within 5 minutes, increase its price by 10%.
- After 10 minutes, the price must reset to its original base_price.

3. Wallet System

Simulate an in-app wallet to handle ticket purchases.

- Requirements:
- Default wallet balance: ₹50,000.
- Deduct the final price upon successful booking.
- If the wallet balance is insufficient, show a clear validation error.

4. Ticket PDF Generation

Generate a downloadable PDF ticket after every successful booking.

- The PDF must include:
- Passenger name
- Airline & Flight ID
- Route (Departure → Arrival)
- Final price paid

- Booking date & time
- Unique PNR

5. Booking History Page

Create a page displaying complete booking history.

- Each booking must show:
- Flight details
- Amount paid
- Booking date
- PNR
- Button to download the ticket again

Bookings may be stored either in the database (preferred) or in browser local storage (acceptable).

Optional Enhancements (Bonus Points)

- Sorting & filtering flights
- Surge pricing indicators / countdown timers
- Responsive UI with TailwindCSS / Bootstrap
- Basic authentication (login/register)
- Dockerized setup
- Clean and meaningful Git commit history
- Search by departure/arrival cities

Submission Requirements

- GitHub Repository URL
- README with clear setup & run instructions
- Short demo video (optional)
- Live deployment link (optional but preferred)

Recommended Technology Stack (Optional)

- Frontend: React / Next.js / Angular
- Backend: Node.js (Express) / Python (FastAPI)
- Database: MongoDB / MySQL / PostgreSQL
- UI: TailwindCSS / Bootstrap
- State Management: Redux / Context API / NgRx

Evaluation Criteria

- Code quality & project structure
- UI/UX & overall presentation
- Effective use of the database

- Correct implementation of dynamic pricing
- Wallet functionality & validations
- PDF ticket generation
- Error handling
- Clarity of README
- Bonus enhancements

Objective

Build a clean, functional, and production-ready flight booking application that demonstrates your technical depth, creativity, and ability to deliver real features end-to-end.

Good luck — build something exceptional!